

TASU2: Controllable CTC Simulation for Alignment and Low-Resource Adaptation of Speech LLMs



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1 Introduction

- Speech LLM post-training relies on **large-scale paired audio–text data**, which is costly to collect.
- TASU** [1] enables **text-only alignment** by simulating CTC posteriors from transcripts, but its random simulation gives **little control over uncertainty and error rate** — curricula stay heuristic.
- In **low-resource domains**, audio fine-tuning can degrade the source domain; plain-text adaptation [2] mismatches the acoustic decoding interface and gains less than TTS [3].
- TASU2** simulates CTC posteriors **under a specified WER range**: closer to real acoustic posteriors, controllable, and TTS-free.

Key results at a glance

12.12%

Medical WER, text-only Stage 2
best; TTS 12.79 · raw audio 12.35

+0.02/+0.07

LibriSpeech WER shift (clean/other)
after target adaptation

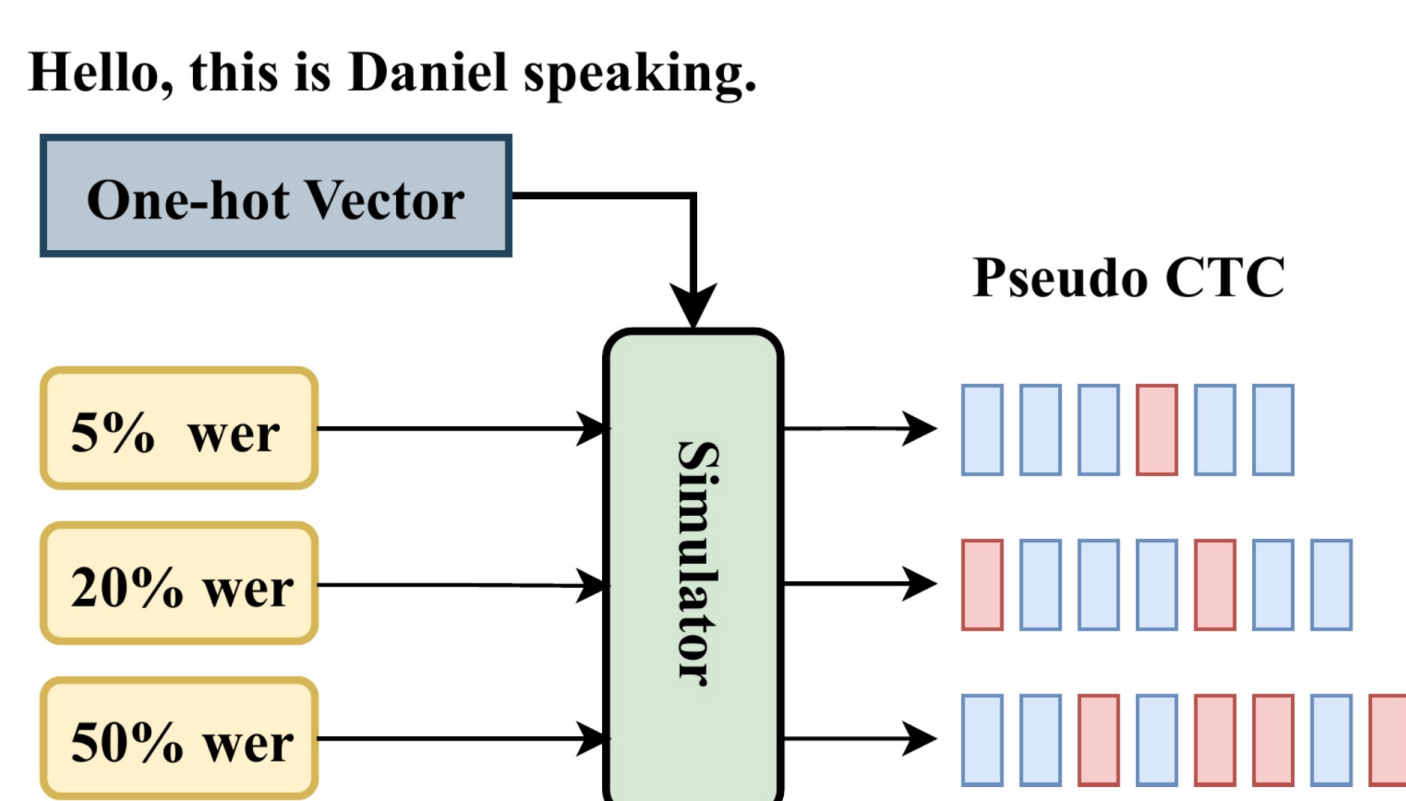
2.34 → 1.05

KL to real teacher CTC posteriors
TASU-style simulation → TASU2

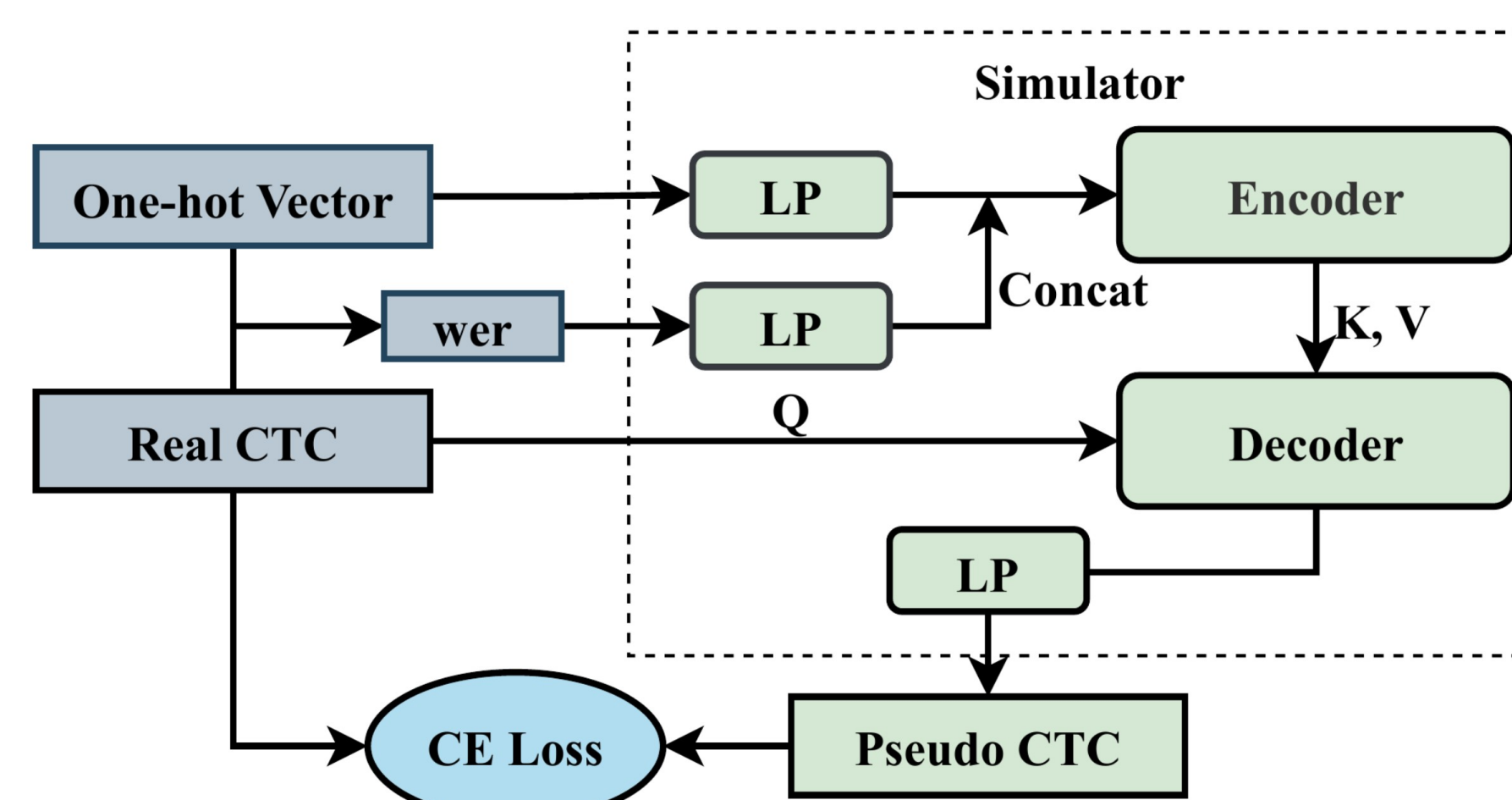
8.15%

LibriSpeech test-other, text only
TASU 9.90 · SLAM-CTC (audio) 8.59

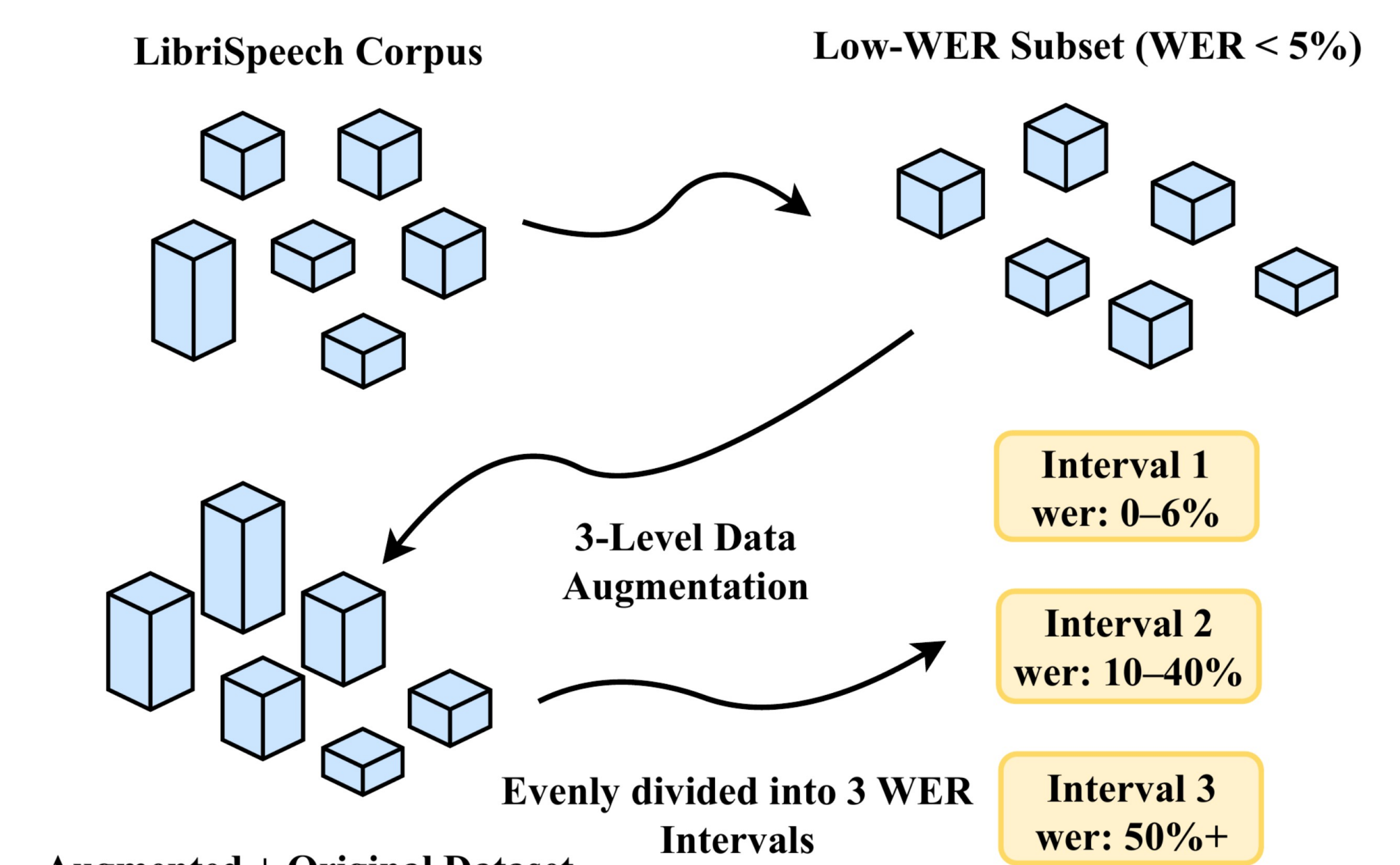
2 TASU2: WER-controllable text-to-CTC posterior simulation



a. The function of Simulator



b. The Structure of Simulator and Training details



c. Simulator Training Data Construction

Figure 1. Overview of TASU2 — (a) function of the simulator, (b) simulator structure and training, (c) simulator training data construction.

(a) What the simulator does

- Transcript y + WER code $c \rightarrow$ pseudo CTC posterior frames, each a distribution over the vocabulary including blank.
- Replaces TASU’s random label smoothing, deletions and insertions with a learned, WER-conditioned generator.

(b) Architecture & training signal

- Transformer encoder–decoder (6 + 6 layers, $d = 512$) + learned WER embedding; one frame per step.
- Posterior-level CE against teacher CTC posteriors:

$$L_{\text{sim}}(\theta) = E\left[-\frac{1}{\sum_i m_i} \sum_i m_i \sum_v p_{i,v} \log \hat{p}_{i,v}\right]$$

(c) WER-conditioned training data

- 1/7 of LibriSpeech training dataset** + noise / reverb variants $\rightarrow$ teacher ASR posteriors and greedy-hypothesis WER $\rightarrow$ code c :

bin 1 · 0–6% bin 2 · 10–40% bin 3 · 50–150%

Discrete bins resist measurement noise and skewed WER distributions.

3 Is the simulator faithful and controllable?

Method	CE ↓	KL ↓	Acc ↑	ProbDiff ↓
TASU-style baseline	2.5149	2.3372	0.8220	0.0813
TASU2 (AR simulator)	1.2296	1.0497	0.8782	0.0615

Table 1. Similarity to teacher CTC posteriors (unconditioned); lower is better except Acc.

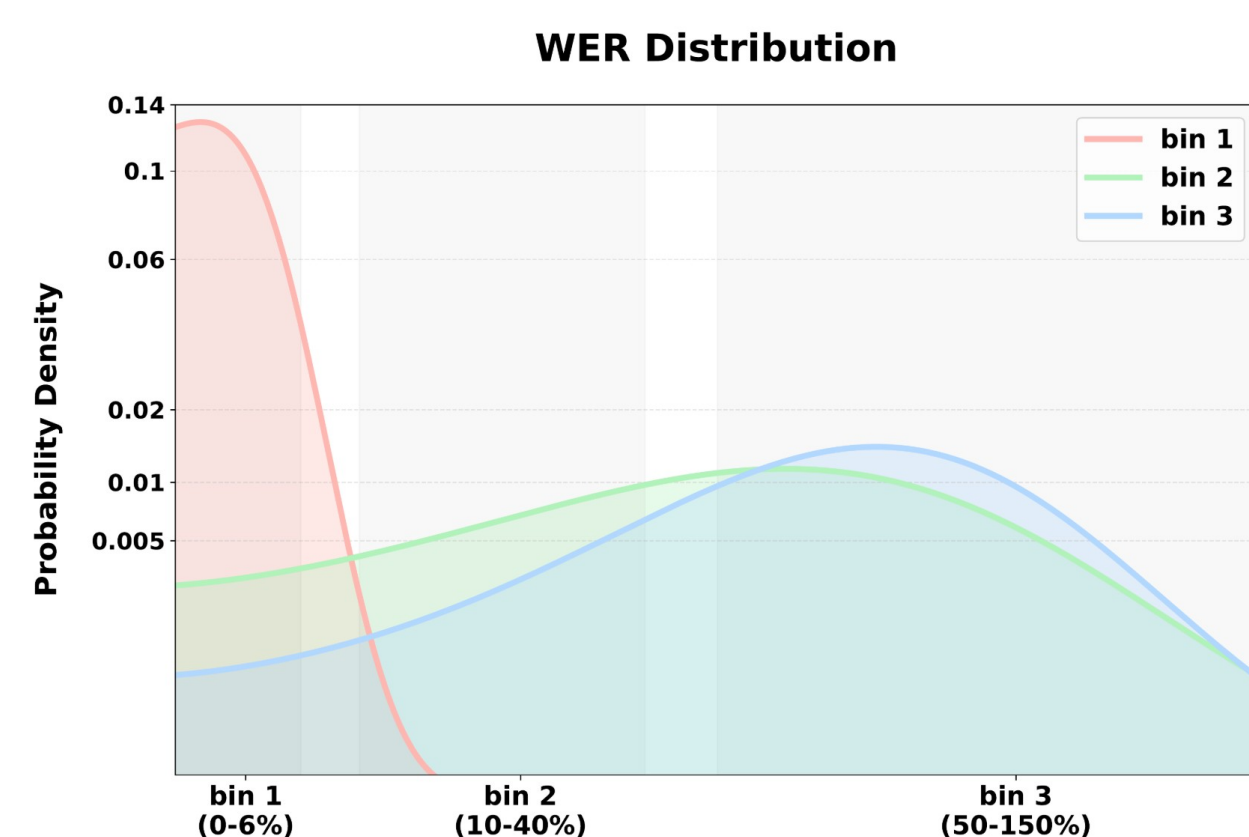


Figure 2. Realized WER of simulated posteriors (fixed decoder) under each WER-bin code.

- Closer to real CTC**: CE and KL roughly halve vs. TASU-style simulation; argmax agreement $0.822 \rightarrow 0.878$.
- Controllable**: each bin code yields a distinct error regime that tracks its target interval.

4 Alignment and generalization without audio

System	Training data	LibriSpeech clean / other	SlideSpeech	TED-LIUM 3
TASU	Libri · text	4.57 / 9.90	24.07	19.36
TASU	Libri + Slide · text	4.21 / 10.31	18.70	13.23
SLAM-CTC	Libri · audio–text	3.13 / 8.59	18.59	14.61
TASU2 uncond.	Libri · text	4.63 / 9.82	16.41	14.02
TASU2 bin 1	Libri · text	3.41 / 8.15	17.67	14.49
TASU2 bin 1	Libri + Slide · text	3.94 / 8.25	17.31	13.93

Table 2. WER (%) ↓ on LibriSpeech (in-domain), SlideSpeech and TED-LIUM 3 (out-of-domain). uncond. = no WER code; bin 1 = 0–6% WER code.

- With LibriSpeech text only, **TASU2 (bin 1) beats TASU on every test set**; 8.15 on test-other is below audio-trained SLAM-CTC (8.59).
- Unconditioned simulation transfers best to SlideSpeech but regresses in-domain; **WER binning balances both**.
- Multi-task model, 0 h training audio: LibriSpeech 3.68 / 8.25 vs. TASU 6.47 / 10.35; CoVoST2 En $\rightarrow$ Zh BLEU 33.08 vs. 33.35.

5 Low-resource domain adaptation

KEY RESULT

System	Stage 2 (target data)	LibriSpeech clean / other	Medical test
SLAM-CTC Stage 1: audio	—	2.43 / 6.07	17.55
	Raw text [2]	2.56 / 6.62	13.62
	TTS audio	2.72 / 6.80	12.79
	Raw audio	2.70 / 6.76	12.35
TASU2 bin 1 Stage 1: text Sim-CTC	—	2.94 / 7.16	15.34
	Text Sim-CTC	2.96 / 7.23	12.12

Table 3. Two-stage adaptation LibriSpeech $\rightarrow$ Medical, WER (%) ↓. Stage 1 on the source domain, Stage 2 on the low-resource target. LoRA ($r = 16$, $\alpha = 32$) for all systems.

- Best Medical WER with text-only Stage 2: 12.12** — ahead of text-only fine-tuning (13.62), TTS augmentation (12.79) and raw target audio (12.35).
- Source domain preserved**: LibriSpeech shifts only +0.02 / +0.07, vs. +0.27 / +0.69 for SLAM-CTC with raw audio (whose audio-trained Stage 1 starts lower).

6 Conclusion

- TASU2: a **WER-controllable text-to-CTC simulator** trained with distribution-level supervision; its posteriors better match acoustic CTC behaviour.
- Text-only post-training with TASU2 improves **alignment and cross-domain generalization** over TASU.
- In low-resource targets it **beats text-only and TTS-based adaptation** while retaining the source domain — practical when paired audio is scarce.

Setup Speech LLM as in TASU: SenseVoice-Small encoder [4] $\rightarrow$ Linear–SiLU–Linear projector $\rightarrow$ Qwen2.5-1.5B [5]. Stage 1: LR 5×10^{-5} , 5 epochs; AdamW, DeepSpeed ZeRO-2, 8 $\times$ Ascend 910B. Main results condition on bin 1 (0–6%). Test sets: LibriSpeech test-clean / other, Medical (8 h), TED-LIUM 3, SlideSpeech, CoVoST2 En $\rightarrow$ Zh.

References [1] J. Peng et al., “TASU: Text-only alignment for speech understanding,” arXiv:2511.03310, 2026. [2] Y. Fang et al., “Low-resource domain adaptation for speech LLMs via text-only fine-tuning,” arXiv:2506.05671, 2025. [3] E. Casanova et al., “ASR data augmentation in low-resource settings using cross-lingual multi-speaker TTS and cross-lingual voice conversion,” 2023. [4] K. An et al., “FunAudioLLM: Voice understanding and generation foundation models for natural interaction between humans and LLMs,” 2024. [5] Qwen Team, “Qwen2.5 technical report,” 2025.

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